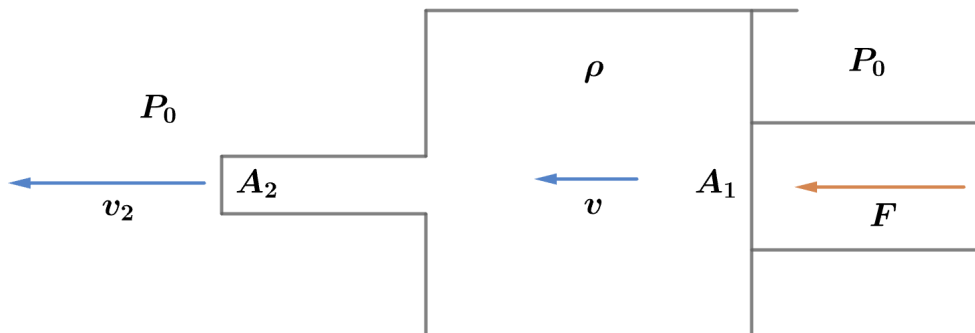


# 2024 F=ma Exam: Problem 21

Kevin S. Huang



We conserve energy by Bernoulli's principle,

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2$$

applied to the two ends of the syringe. In our case,

$$\begin{aligned} y_1 &= y_2 \\ P_1 &= P_0 + \frac{F}{A_1} \\ P_2 &= P_0 \end{aligned}$$

so we have

$$\begin{aligned} P_0 + \frac{F}{A_1} + \frac{1}{2}\rho v^2 &= P_0 + \frac{1}{2}\rho v_2^2 \\ F &= \frac{A_1}{2}\rho(v_2^2 - v^2) \end{aligned}$$

By the continuity equation,

$$\begin{aligned} A_1 v &= A_2 v_2 \\ v_2 &= \frac{A_1 v}{A_2} \end{aligned}$$

Substituting into our expression for  $F$ ,

$$F = \frac{A_1}{2}\rho \left( \frac{A_1^2}{A_2^2} - 1 \right) v^2 \approx \frac{A_1^3 \rho v^2}{2A_2^2}$$

using the assumption that  $A_1 \gg A_2$ . Thus, the answer is D.